SIMATS ENGINEERING

CSA6303- Threat Intelligence and Network Security A1-Malware Analysis Assignment

Assessment Questions

Subject	Threat Intelligence	**Total Marks**	50
Questions	10 (all compulsory)	**Marks per Question**	5
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Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.

CO3: Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

Code of Conduct:

I N.Harshith (192311129) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

N.Harshith
Signature of Student

Questions :

1. Malware Analysis Using Static Techniques

Static malware analysis is the process of examining a suspicious file without executing it. This helps analysts identify potential threats while avoiding the risk of infection.

Steps:

1.**Calculate File Hashes**

- Generate MD5, SHA-1, or SHA-256 hashes.
- Compare hashes with VirusTotal, MalwareBazaar, or threat intelligence databases.
- Matching hashes may indicate known malware.

2. **Inspect File Properties**

- Check file size, compilation date, digital signature, and metadata.
- Suspicious indicators include missing signatures, fake publishers, or unusual timestamps.

3. **Extract Strings**

- Use tools like Strings, FLOSS, or BinText.
- Look for:
    - URLs
    - IP addresses
    - Registry keys
    - File paths
    - Commands

4. **Analyze Imports**

- Examine imported Windows API functions.
- Functions such as:
    - CreateRemoteThread()
    - WriteProcessMemory()
    - WinExec()
    - URLDownloadToFile() may indicate malicious behavior.

5. **PE Header Analysis**

- Review Portable Executable (PE) structure.
- Detect packers, obfuscation, or unusual sections.

Benefits

- Safe analysis method.
- Identifies suspicious characteristics quickly.
- Helps estimate malware capabilities before execution.

Risk Assessment

Finding suspicious imports, encoded strings, known malicious hashes, or packing techniques suggests a high probability that the executable is malicious.

2. Malware Analysis Using Dynamic Techniques

Dynamic analysis involves executing malware in a controlled environment to observe its behavior.

Analysis Environment

- Isolated Virtual Machine (VM)
- Sandbox (Any.Run, Cuckoo Sandbox)
- No connection to production systems

Investigation Steps

1. Create a clean VM snapshot.
2. Install monitoring tools:
    - Process Monitor
    - Process Explorer
    - Regshot
    - Wireshark
3. Execute the suspicious file.
4. Observe:
    - New processes
    - Registry modifications
    - File creation/deletion
    - Network connections

Indicators of Compromise (IoCs)

Dynamic analysis may reveal:

- Suspicious domains
- Malicious IP addresses
- Dropped files
- Persistence mechanisms
- Registry changes
- Scheduled tasks

Outcome

The analyst can determine actual malware behavior and gather evidence used for detection and incident response.

3. Malware Investigation Using a Threat Intelligence Platform

Threat intelligence platforms provide contextual information about malware and known threats.

1. Obtain file hash.
2. Search the hash in:

- VirusTotal
- IBM X-Force Exchange
- AlienVault OTX
- MalwareBazaar

3. Review:

- Malware family name
- Detection rates
- Known IoCs
- Associated campaigns

Assessing Reliability

Verify information using:

- Multiple reputable intelligence sources
- Vendor reputation
- Confidence scores
- Recent threat reports

Containment Measures

If the file belongs to a known malware family:

- Isolate affected hosts.
- Block malicious domains and IPs.
- Remove malicious files.
- Reset passwords if credential theft is suspected.
- Scan all systems for related IoCs.

Result

Threat intelligence provides valuable context and accelerates incident response activities.

4. Malware Detection Through Data Collection and Correlation

When multiple systems exhibit suspicious behavior, analysts correlate data from various sources.

Data Sources

Endpoint Logs

- Process creation events
- File modifications
- User activity

DNS Logs

- Suspicious domain lookups
- Domain Generation Algorithm (DGA) activity

Firewall Logs

- Outbound connections
- Blocked traffic

File Hashes

- Presence of identical suspicious files

Correlation Process

1. Identify suspicious process names.
2. Match processes on multiple systems.
3. Check DNS records for common domains.
4. Review firewall logs for shared outbound destinations.
5. Compare file hashes across devices.

Example :

Computer A:

- Process: updater.exe
- Domain: maliciousxyz.com

Computer B:

- Same process
- Same domain
- Same file hash

These correlations strongly indicate malware infection.

Outcome

Combining data sources increases confidence and helps identify infection scope across the university network.

5. Malware Investigation Using SIEM Tools

SIEM solutions such as Splunk, QRadar, Microsoft Sentinel, and ArcSight centralize logs and correlate events.

Scenario Analysis

Alert Sources:

- Suspicious process execution
- Failed login attempts
- Abnormal network connections

Investigation Steps

Event Correlation

The SIEM links:

- Same host

- Same timestamp range
- Same user account

Timeline Creation

Create a sequence:

1. User receives file.
2. Process executes.
3. Failed logins occur.
4. Network communication begins.

Threat Validation

Investigate whether:

- Account compromise occurred.
- Malware established persistence.
- Data exfiltration happened.

Prioritization Factors

- Critical systems affected
- Multiple alerts correlated
- Confirmed malicious connections
- Sensitive data exposure

Outcome

SIEM correlation reduces analysis time and provides a complete attack timeline.

6. Malware Detection and Log Analysis Using Splunk

Splunk allows centralized log collection, searching, and correlation.

Relevant Splunk Searches

Suspicious External Connections

Plain Text
spl isn’t fully supported. Syntax highlighting is based on Plain Text.

```
index=firewall action=allowed
| stats count by src_ip dest_ip
```

DNS Queries

Plain Text
spl isn’t fully supported. Syntax highlighting is based on Plain Text.
index=dns
| stats count by query

Process Execution

Plain Text
spl isn’t fully supported. Syntax highlighting is based on Plain Text.
index=endpoint
| search process_name="*.exe"

Indicators to Look For

- Beaconing traffic
- Frequent connections to one IP
- Rare domains
- Suspicious executable launches

Event Correlation

Combine:

- Process logs
- DNS logs
- Network activity

Example:

1. malware.exe starts.
2. DNS resolves bad-domain.com.
3. Host connects to remote IP.

Evidence Documentation

Record:

- Timestamps
- Hostnames
- Domains
- IP addresses
- File hashes
- User accounts

Outcome

Splunk provides visibility into malware communication and supports forensic investigations.

7. Malware Investigation Using the ELK Stack

The ELK Stack consists of:

- **Elasticsearch**: Storage and search
- **Logstash**: Data ingestion and processing
- **Kibana**: Visualization and dashboards

Investigation Workflow

Collection

Logstash gathers:

- Web server logs
- Firewall logs
- Endpoint telemetry

Processing

Normalize logs into a common format.

Searching

Use Elasticsearch to search for:

- Suspicious domains
- Malicious IP addresses
- Specific file hashes

Visualization

Kibana dashboards can display:

- Connection trends
- DNS requests
- Malware-related events
- Host activity

Correlation

Link:

- Endpoint alerts
- File modifications
- Network traffic
- User activity

Benefits

- Centralized visibility
- Fast searching
- Real-time monitoring
- Visual investigation support

8. Threat Hunting for Malware Activities

Threat hunting is a proactive process used to discover hidden threats that automated tools may miss.

Investigation Approach
Step 1: Examine Running Processes

Check:

- Unknown executables
- High CPU utilization
- Unsigned programs

Step 2: Review Network Activity

Identify:

- External IP connections
- Repeated outbound traffic
- Unusual ports

Step 3: Check Persistence

Search for:

- Scheduled tasks
- Registry Run keys
- Startup folder entries

Step 4: Examine Files

Collect:

- File hashes
- Creation timestamps
- Locations

Step 5: Gather IoCs

Examples:

- File hashes
- Domains
- IP addresses
- Process names
- Registry keys

Remediation

- Isolate workstation.
- Remove malware.
- Block IoCs.
- Reset credentials.
- Reimage system if necessary.
- Monitor for re-infection.

Outcome

Threat hunting uncovers stealthy malware that may evade traditional security controls.

9. Comparing Malware Analysis Tools Used in the IT Industry

Tool Category	Purpose	Advantages	Limitations
Static Analysis Tools	Examine file without execution	Safe and fast	Cannot observe real behavior
Dynamic Analysis Tools	Run malware in sandbox	Reveals actual behavior	Malware may detect sandbox
Network Traffic Analyzers	Analyze communications	Detect C2 traffic	Limited file insight
Threat	Provide threat context	Known IoCs and	Depend on

Intelligence Platforms		attribution	available intelligence

Examples

Static Analysis

- PEStudio
- Detect It Easy (DIE)
- Strings
- Ghidra
- IDA Free

Dynamic Analysis

- Cuckoo Sandbox
- Any.Run
- Procmon
- Process Explorer

Network Analysis

- Wireshark
- Zeek
- TCPDump

Threat Intelligence

- VirusTotal
- AlienVault OTX
- MalwareBazaar
- AbuseIPDB

Recommended Investigation Workflow

1. Static analysis.
2. Threat intelligence lookup.
3. Dynamic sandbox execution.
4. Network traffic inspection.
5. IoC extraction and validation.

This layered approach provides the most complete understanding of malware behavior.

10. Malware Analysis Report and Incident Response

Malware Analysis Report Template

1. Executive Summary

- Brief description of incident
- Severity rating
- Affected systems

2. File Information

Field	Value
File Name	suspicious.exe
File Type	PE Executable
MD5	xxx
SHA256	xxx
File Size	xxx
Detection Rate	xx/yy

3. Analysis Methodology

Static Analysis

- Hash verification
- String extraction
- Import analysis
- PE header inspection

Dynamic Analysis

- Sandbox execution
- Process monitoring
- Registry monitoring
- Network monitoring

4. Observed Behavior

- Created new files
- Modified registry keys
- Established persistence
- Contacted external domains
- Spawned child processes

5. Indicators of Compromise (IoCs)

File Hashes

- SHA256: XXXXX

Domains

- malicious-domain.com

IP Addresses

- 192.168.x.x

Registry Keys

- HKCU\Software\Microsoft\Windows\CurrentVersion\Run

File Paths

- C:\Users\Public\malware.exe

6. Threat Intelligence Findings

- Malware family identified
- Detection sources
- Historical activity
- Associated campaigns

7. Risk Assessment

Area	Impact
Confidentiality	High
Integrity	High
Availability	Medium

8. Containment Actions

- Isolate host.
- Block domains.
- Block IPs.
- Disable compromised accounts.

9. Recovery Actions

- Remove malware.
- Restore clean backups.
- Patch vulnerabilities.
- Reset credentials.
- Validate system integrity.

10. Lessons Learned

- Improve email filtering.
- Strengthen endpoint monitoring.
- Conduct security awareness training.
- Update incident response procedures.

Conclusion

The investigation confirms malicious activity through suspicious processes, system modifications, and communication with external infrastructure. Immediate containment, eradication, and recovery actions should be implemented to prevent further compromise and reduce organizational risk.